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VEHICLE CAPTURE ASSEMBLIES AND RELATED DEVICES, SYSTEMS, AND METHODS

Abstract

Vehicle capture assemblies and related devices, systems, and methods include one or more probe assemblies for passively engaging with and securing the target vehicle.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a divisional of U.S. patent application Ser. No. 17/207,646, filed Mar. 20, 2021, for “VEHICLE CAPTURE ASSEMBLIES AND RELATED DEVICES, SYSTEMS, AND METHODS,” which application claims the benefit of the filing date of U.S. Provisional Patent Application Ser. No. 63/019,923, filed May 4, 2020, for “VEHICLE CAPTURE ASSEMBLIES AND RELATED DEVICES, SYSTEMS, AND METHODS,” the disclosure of each of which is incorporated herein in its entirety by reference.

TECHNICAL FIELD

[0002] The present disclosure relates to systems, devices, assemblies, apparatus, and methods for vehicle (e.g., spacecraft) docking. In some embodiments, the present disclosure includes a primarily passive vehicle capture assembly for engaging with an associated vehicle and related devices, systems, and methods.

BACKGROUND

[0003] Docking assemblies and devices may be utilized to mechanically connect two or more vehicles (e.g., spacecraft) to each other. Such spacecraft may be vehicles (e.g., self-propelled vehicles) designed for short-term space flights and/or may be configured to remain in space for a long period of time. The spacecraft may be intended to perform a specific function in a space mission, such as supplying resources to and/or altering the orbit of a target vehicle. In some instances, the spacecraft may be a space station, satellite, or another suitable structure.

[0004] The connection of two or more spacecraft may enable the transfer of resources from one spacecraft to another spacecraft. For example, a spacecraft may dock with a space station to deliver crew and resources. In another example, a spacecraft may dock with a satellite to perform maintenance and repair of one or more components of that satellite. In yet an additional example, a spacecraft may dock with another vehicle to provide a specific mission function, such as, for example, propulsion for a descent to or an ascent from an astronomical body or to transfer to a select location for the mission.

[0005] Conceptualized methods of docking to spacecraft consist of complex mechanical implements. Various patents and publications have considered such methods, including U.S. Pat. Nos. 3,508,723, 4,018,409, 4,177,964, 4,219,171, 4,391,423, 4,588,150, 4,664,344, 4,898,348, 5,005,786, 5,040,749, 5,094,410, 5,299,764, 5,364,046, 5,372,340, 5,490,075, 5,511,748, 5,735,488, 5,803,407, 5,806,802, 6,017,000, 6,299,107, 6,330,987, 6,484,973, 6,523,784, 6,742,745, 6,843,446, 6,945,500, 6,969,030, 7,070,151, 7,104,505, 7,207,525, 7,216,833, 7,216,834, 7,240,879, 7,293,743, 7,370,834, 7,438,264, 7,461,818, 7,484,690, 7,513,459, 7,513,460, 7,575,199, 7,588,213, 7,611,096, 7,611,097, 7,624,950, 7,815,149, 7,823,837, 7,828,249, 7,857,261, 7,861,974, 7,861,975, 7,992,824, 8,006,937, 8,006,938, 8,016,242, 8,033,508, 8,056,864, 8,074,935, 8,181,911, 8,196,870, 8,205,838, 8,240,613, 8,245,370, 8,333,347, 8,412,391, 8,448,904, 8,899,527, 9,108,747, 9,302,793, 9,321,175, and 9,399,295; U.S. Patent Application Pub. Nos. 2004/0026571, 2006/0145024, 2006/0151671, 2007/0228220, 2009/0001221, 2012/0112009, 2012/0325972, 2013/0103193, 2015/0008290, 2015/0314893, 2016/0039543, and 2016/0039544; European Patent Nos. EP 0092602 A1, EP 0541052, 0741655 B1, 0741655 B2, and 1654159; PCT Pub. Nos. 2005/110847, 2005/118394, 2014/024,199, and 2016/030890; Japan Patent Nos. JPH01282098 and JPH01226497; *Automated Rendezvous and Docking of Spacecraft*, Fehse, Wigbert, Cambridge University Press (2003); *On-Orbit Servicing Missions: Challenges and Solutions for Spacecraft Operations*, Sellmaier, F., et al., SpaceOps 2010 Conference, AIAA 2010-2159 (2010); and *Towards a standardized grasping and refueling on-orbit servicing for geo spacecraft*, Medina, Alberto, et al., Acta Astronautica vol. 134, pp. 1-10 (2017); DEOS-The In-Flight Technology Demonstration of German's Robotics Approach to Dispose Malfunctioned Satellites, Reintsema, D., et al., the disclosure of each of which is hereby incorporated herein in its entirety by this reference.

[0006] However, mechanical complexity and/or active control complexity that is present in many of the above designs increases the likelihood of component failure, which can result in failure in the docking and maintenance process.

BRIEF SUMMARY

[0007] Some embodiments of the disclosure may include a vehicle capture assembly comprising a probe assembly including one or more retention elements for engaging with and securing a target vehicle and an extendable lance coupled to the probe assembly at a distal portion of the lance. The probe assembly may be configured to passively engage with and secure the target vehicle and to passively disengage from and release the target spacecraft.

[0008] Some embodiments of the disclosure may include a vehicle capture assembly comprising a probe assembly including one or more retention elements for engaging with and securing a target vehicle in a deployed position. The probe assembly may be configured to passively engage with and secure the target vehicle with the one or more retention elements in the deployed position. The vehicle capture assembly may include a biasing element for biasing the one or more retention elements in the deployed position. The one or more retention elements may be configured to be moved toward a stowed position against a force of the biasing element in response to a force applied to the one or more retention elements. The vehicle capture assembly may include an extendable lance coupled to the probe assembly where the probe assembly is positioned at a distal portion of the lance and an actuation element coupled to the one or more retention elements by one or more linkages. The actuation element may be for interacting with the biasing element to bias the one or more retention elements in the deployed position and to return the one or more retention elements to the deployed position after being forced into the stowed position.

[0009] Some embodiments of the disclosure may include a spacecraft capture system comprising two or more vehicle capture assemblies. The vehicle capture assemblies each include a probe assembly including one or more retention elements for engaging with and securing a target spacecraft and an extendable lance coupled to the probe assembly at a distal portion of the lance. The probe assembly is configured to engage with a portion of the target spacecraft. The two or more vehicle capture assemblies may be configured to substantially simultaneously retract each respective probe assembly of the two or more vehicle capture assemblies in order to secure the target spacecraft.

[0010] Some embodiments of the disclosure may include a method of capturing a spacecraft including extending a lance of a vehicle capture assembly toward a target spacecraft, passively engaging a probe of the vehicle capture assembly with the target spacecraft, the probe being coupled to the lance, retracting the lance of the vehicle capture assembly to at least partially secure the target spacecraft, and passively releasing one or more barbs of the probe from engagement with the target spacecraft.

[0011] Some embodiments of the disclosure may include a method of capturing a target spacecraft including biasing one or more barbs of a probe of a vehicle capture assembly in a deployed position with an actuation element coupled to the one or more barbs, inserting the one or more barbs into a docking element of the target spacecraft, at least partially retracting the one or more barbs to a retracted position in response to a force applied to the one or more barbs by the target spacecraft, moving the one or more barbs back to the deployed position to secure the one or more barbs to the docking element with the actuation element, moving the actuation element along the probe against a biasing force, and, in response to moving the actuation element, moving the one or more barbs to a release position in order to disengage the one or more barbs from the docking element of the target spacecraft.

[0012] Some embodiments of the disclosure may include one or more probe assemblies that are configured to be received in a respective structure (e.g., one or more capture cones) on a target vehicle. Retention features on the one or more probe assemblies may enable movement of the probe assembly to provide an approximate universal joint between the one or more probe

assemblies and the target vehicle.

[0013] The above summary is not intended to describe each illustrated embodiment or every implementation of the present disclosure.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0014] The drawings included in the present application are incorporated into, and form part of, the specification. They illustrate embodiments of the present disclosure and, along with the description, serve to explain the principles of the disclosure. The drawings are only illustrative of certain embodiments and do not limit the disclosure.

[0015] FIG. 1 is a schematic side view of a capture vehicle with a vehicle capture assembly and a target vehicle according to one or more embodiments of the disclosure.

[0016] FIG. 2 is an isometric view of a vehicle capture assembly in an initial position according to one or more embodiments of the disclosure.

[0017] FIG. 3 is a partial cross-sectional side view of a vehicle capture assembly in an initial position according to one or more embodiments of the disclosure.

[0018] FIG. 4 is an isometric view of a vehicle capture assembly in an extended or extending position according to one or more embodiments of the disclosure.

[0019] FIG. 5 is an isometric view of a docking assembly according to one or more embodiments of the disclosure.

[0020] FIG. 6 is an isometric view of a vehicle capture system including multiple vehicle capture assemblies according to one or more embodiments of the disclosure.

[0021] FIG. 7 is a partial cross-sectional side view of a vehicle capture assembly in an initial position according to one or more embodiments of the disclosure.

[0022] FIG. 8 is a partial cross-sectional side view of a vehicle capture assembly being received in a docking assembly according to one or more embodiments of the disclosure.

[0023] FIG. 9 is a partial cross-sectional side view of a vehicle capture assembly received in a docking assembly according to one or more embodiments of the disclosure.

[0024] While the disclosure is amenable to various modifications and alternate forms, specifics thereof have been shown by way of example in the drawings and will be described in detail. It should be understood, however, that the intention is not to limit the disclosure to the particular embodiments described. On the contrary, the intention is to cover all modifications, equivalents, and alternatives falling within the scope of the disclosure.

DETAILED DESCRIPTION

[0025] As used herein, the term “substantially” in reference to a given parameter means and includes to a degree that one skilled in the art would understand that the given parameter, property, or condition is met with a small degree of variance, such as within acceptable manufacturing tolerances. For example, a parameter that is substantially met may be at least about 90% met, at least about 95% met, at least about 99% met, or even 100% met.

[0026] Embodiments of the present disclosure may include capture and mating assemblies and systems used for mechanical docking of two or more vehicles (e.g., spacecraft in orbit, with or without self-propulsion), using one or more vehicle capture assemblies (e.g., two or more, three, four, etc.) on the capture vehicle. The vehicle capture assemblies are configured to engage with a docking portion of a target vehicle to be captured (e.g., one or more docking cones coupled to the target vehicle). Some embodiments may enable the autonomous capture and docking of spacecraft with relatively large mass and inertia, while incurring minimal disturbance to either vehicle. Some embodiments provide benefits in the form of a relatively simple docking architecture with compliancy for improved reliability and safety, that is, preventing damage to the spacecraft.

[0027] Some embodiments may reduce mechanical complexity of the docking apparatus by removing the need for an actively driven probe assembly for the docking process. For example, extension and/or retraction of retention elements of the probe assembly to capture and/or release the target vehicle may be effected substantially without the use of active components or device within the probe (e.g., a motor or other electronic and/or hydraulic actuation assemblies, such as a solenoid). Such embodiments may reduce or eliminate the need for electronic components (e.g., signal conductors, wiring, power systems, switches, etc.) in portions of the vehicle capture assembly (e.g., in a lance and/or probe of the vehicle capture assembly).

[0028] FIG. 1 depicts a schematic side view in which a capture vehicle **10** (e.g., a servicing vehicle, a chaser spacecraft, a transfer spacecraft, etc.) may be operated to approach, capture, dock to, supply, transfer cargo or resources to, and/or service a target vehicle **11**, according to one or more embodiments of the disclosure.

[0029] Capture vehicle **10** and target vehicle **11** each may be a spacecraft or a satellite situated in orbit around a body. The capture vehicle **10** may be a spacecraft designed to approach, capture, dock to, and undock from the target vehicle **11**. Docking of the capture vehicle **10** to target vehicle **11** may enable a specific function in a space mission. For example, the connection of the vehicles **10**, **11** may enable the transfer of resources (e.g., cargo, equipment, passengers, crew, etc.) from one vehicle to another vehicle, may enable vehicle repair, and/or may enable a specific mission function (e.g., propulsion for a descent to or an ascent from an astronomical body or to transfer to a select location in space for the mission).

[0030] Capture vehicle **10** may be designed to dock with more than one target vehicle **11**. For example, the capture vehicle **10** may be provided with a docking mechanism (e.g., vehicle capture assembly **22**) that enables the capture vehicle **10** to dock and undock from multiple target vehicles **11**. The capture vehicle **10** may be configured to dock with one or more of the target vehicles **11** comprising one or more docking elements **18** (e.g., a docking cone, an engine, etc.).

[0031] In some embodiments, and as discussed below, the docking element **18** may be configured to enable immediate release of the coupling between the vehicles **10**, **11**. For example, the docking element **18** may be coupled to the target vehicle **11** with a releasable coupling or union that may be destructively (e.g., with one or more pyrotechnic fasteners, such as explosive bolts) or nondestructively (e.g., with a releasable union) released from the target vehicle **11** in order to free at least a portion of the docking element **18**.

[0032] As depicted, the capture vehicle **10** may include a spacecraft body **12**, a docking platform **14**, a main thruster **17**, gimbaled thrusters **20**, and the vehicle capture assembly **22**. As noted above, the vehicle capture assembly **22** may include retention elements that directly contact and secure the target vehicle **11** in a manner that does not require the use of active components such as a motor directly actuating the retention elements (e.g., is not required to be driven in an active manner). Rather, the retention elements may use passive methodology or mechanisms, such as mechanical forces (e.g., biasing forces), to engage with the target vehicle **11**.

[0033] Mechanical forces (e.g., biasing forces) may be used to release (e.g., nondestructively release) the retention elements to move the retention elements toward a stowed or disengaged position in order to release the target spacecraft **11** without the use of a motor directly driving the retention elements. Such embodiments may reduce, or even eliminate, the need for electronic components (e.g., signal conductors, electrical wiring, power systems, switches, motor, heaters, thermistors, helical harnesses, etc.) in portions of the vehicle capture assembly **22** (e.g., in a lance and/or probe of the vehicle capture assembly **22**).

[0034] As discussed below, while a motor (e.g., only a single and solitary motor) may be used to actively move (e.g., translate) the vehicle capture assembly **22** toward and/or away from the target vehicle **11**, such a motor may only indirectly contribute to the engagement and/or disengagement of the retention elements. For example, while the motor may place the retention elements in a selected position relative to the target vehicle, a force applied to the vehicle capture assembly **22** may be

utilized to engage and/or disengage the retention elements (e.g., a force overcoming one or more biasing elements of the vehicle capture assembly **22**) in a passive manner that is not actively driven by a motor or an otherwise electronic device.

[0035] Target vehicle **11** may be a spacecraft to be captured by the vehicle capture assembly **22** of the capture vehicle **10**. Target vehicle **11** may be in low earth orbit, medium earth orbit, geosynchronous orbit, beyond geosynchronous orbit, or in another orbit around an astronomical body, for example, such as Earth, the moon, or another planetary body. Target vehicle **11** may include the docking element **18** and a separation ring **19**.

[0036] Vehicle capture assembly **22** of capture vehicle **10** may be configured to capture target vehicle **11** at docking element **18** and to pull target vehicle **11** and capture vehicle **10** together for docking. When docked, one or more portions of the target vehicle **11** and/or vehicle capture assembly **22** may abut and retain the vehicles **10**, **11** together.

[0037] FIG. **2** depicts an isometric view of a vehicle capture assembly **100** in an initial position that may be used with a capture vehicle. In some embodiments, vehicle capture assembly **100** may be similar to, and include similar components and features of, the vehicle capture assembly **22** of the capture vehicle **10** which is depicted schematically in FIG. **1** and discussed above.

[0038] As shown in FIG. **2**, the vehicle capture assembly **100** includes a probe or probe assembly **102** coupled to a lance or lance assembly **104** (e.g., at a distal portion of end of the lance assembly **104**). The probe assembly **102** includes one or more retention features (e.g., barbs **106**) that extend from the probe assembly **102** at a location proximate a probe tip **108**. The barbs **106** may extend in a direction transverse to a length or longitudinal axis of one or more portions of the vehicle capture assembly **100** (e.g., lateral to a length of the lance assembly **104**). As depicted, the rotatable barbs **106** extend laterally outward and in proximal direction toward the lance assembly **104** in order to capture a target vehicle **11** (FIG. **1**).

[0039] In some embodiments, the barbs **106** may be biased (e.g., spring-loaded) in a selected position. For example, the barbs **106** may be in the depicted deployed position where the barbs **106** may couple with a portion of the target vehicle **11** (FIG. **1**). In additional embodiments, the barbs **106** may be biased in a retracted or stowed position.

[0040] Referring to FIGS. **1** and **2**, the lance assembly **104** may include functionality that enables extension and/or retraction of a portion of probe assembly **102** to facilitate docking of capture vehicle **10** with target vehicle **11**. For example, when capture vehicle **10** is positioned proximate to the target vehicle **11**, the probe assembly **102** may be extended to and inserted into the docking element **18** of the target vehicle **11** with the lance assembly **104**. The lance assembly **104** may include a lance boom **110** that is driven by a motor **112** positioned in a housing **115** of the probe assembly **102**. The motor **112** may be used to actively move (e.g., translate) the lance boom **110** toward and/or away from the target vehicle **11**.

[0041] In some embodiments, the motor **112** may only indirectly contribute to the engagement and/or disengagement of the barbs **106**. For example, while the motor **112** may place the barbs **106** in a selected position relative to the target vehicle **11**, force applied to the barbs **106** (e.g., to overcome the biasing force of the barbs **106** into the deployed position) may be applied as the barbs **106** are inserted into the docking element **18** to engage the barbs **106** in a passive manner that is not actively driven by the motor **112**. As discussed below, movement of the probe assembly **102** (e.g., by forcing the probe assembly **102** into the target vehicle **11**) may be used to release the barbs **106** from the target vehicle **11** (e.g., by overcoming the biasing forces of the barbs **106** in a different manner with internal components of the probe assembly **102**).

[0042] The vehicle capture assembly **100** may include another rearward element (e.g., docking cone **114**) for engaging another portion of the target vehicle **11** (e.g., another portion of the docking element **18**). As depicted, docking cone **114** may be biased toward the barbs **106** (e.g., by spring **116**) in order to secure the target vehicle between the barbs **106** and the docking cone **114**.

[0043] The vehicle capture assembly **100** may include a backstop plate **118** for mating with a

portion of the target vehicle **11** (e.g., in the captured position). One or more additional retention elements (e.g., latches **120**) may be coupled (e.g., rotatably coupled) to the backstop plate **118**. Additional embodiments may include a linkage-type latch. In the captured position, the latches **120** may be actuated to engage with the docking element **18** to secure the target vehicle **11**. In some embodiments, after engagement, the latches **120** may comprise a majority of the rigid connection between the vehicles **10**, **11** (e.g., may be the primary connection). For example, the latches **120** may bear a majority of the forces developed between the two vehicles **10**, **11** while the other attachment points (e.g., the docking cone **114** and/or the barbs **106**) are not primarily used or experience a significantly lower amount of force (e.g., by one or more orders of magnitude).

[0044] FIG. **3** is a partial cross-sectional side view of a vehicle capture assembly (e.g., vehicle capture assembly **100**) in an initial position (e.g., after being initially deployed from a stowed state and before being extended toward the target vehicle **11** (FIG. **1**)). As shown in FIG. **3**, the vehicle capture assembly **100** includes an actuation feature (e.g., cam actuator **122**) that may be coupled to the lance assembly **104** (e.g., at a distal end of the lance boom **110**). The cam actuator **122** may move (e.g., translate) relative to one or more portions of the probe assembly **102** (e.g., the probe tip **108**). For example, the cam actuator **122** may translate relative to (e.g., slide within) the probe tip **108**. In some embodiments, the probe tip **108** may move relative to the overall vehicle capture assembly **100** while the cam actuator **122** is held stationary relative to one or more portions of the vehicle capture assembly **100** (e.g., the lance assembly **104**).

[0045] A biasing element **124** (e.g., a spring) may be positioned between the cam actuator **122** and the probe tip **108** to bias the cam actuator **122** and/or the probe tip **108** in a selected position. For example, the biasing element **124** may bias the probe tip **108** in a position where the cam actuator **122** is separated from the barbs **106**. When a force (e.g., force applied to the probe tip **108** between the vehicles **10**, **11** (FIG. **1**)) overcomes the biasing element **124**, the probe tip **108** may move relative to the cam actuator **122**. A cam tip **126** of the cam actuator **122** may slide through the barbs **106** to move (e.g., rotate) the barbs **106** to another position (e.g., a stowed or released position). Once the force is removed from the probe tip **108**, the biasing element **124** may return the probe tip **108** to an extended position where the barbs **106** may be returned to a deployed or capture position.

[0046] FIG. **4** is an isometric view of a vehicle capture assembly (e.g., vehicle capture assembly **100**) in an extended or extending position. FIG. **5** is an isometric view of a docking assembly **200** for a target vehicle **11** (FIG. **1**).

[0047] Referring to FIG. **1** through **5**, in operation, the vehicle capture assembly **100** may be positioned in the initial state shown in FIGS. **2** and **3**. In some embodiments, the vehicle capture assembly **100** may move to the initial state from a stowed state (e.g., used during transport). In the stowed state, the probe tip **108** and barbs **106** may position (e.g., compress) the probe tip **108** and docking cone **114** toward or in contact with the backstop plate **118**.

[0048] From the initial position, the lance motor **112** may translate the probe assembly **102** toward the target vehicle **11** (e.g., toward the docking assembly **200**) to the extended or extending position shown in FIG. **4**. The lance motor **112** and/or the capture vehicle **10** may move (e.g., force) the probe tip **108** through an outer cone **202** of the docking assembly **200**, through a necked portion **204**, and into an inner volume **206** of the docking assembly **200**. As the probe tip **108** passes through the necked portion **204**, biasing force of the barbs **106** may be overcome by the force of the insertion. The barbs **106** may at least partially retract to pass through the necked portion **204** and may be returned to a deployed or expanded position once in the inner volume **206** to initially capture the target vehicle **11**.

[0049] In some embodiments, where multiple vehicle capture assemblies **100** are utilized (e.g., three vehicle capture assemblies **100**), the combination of the vehicle capture assemblies **100** coupled to respective docking assemblies **18** of the target vehicle **11** may approximate a universal joint or rotational joint to provide for allowance and/or attenuation of movement and/or forces between the vehicles **10**, **11**. For example, both the barbs **106** (e.g., three barbs) and an inner

portion the docking assembly **200** with which a distal end of the barbs **106** engage may comprise complementary surfaces. In some embodiments, both the barbs **106** and the inner portion of the docking assembly **200** may define at least partially spherical surfaces that enable the barbs **106** to move (e.g., slide along) an inner surface of the docking assembly **200** to provide the approximate universal joint.

[0050] In some embodiments, each docking assembly (e.g., three docking assemblies) may provide the combination of a universal joint, a prismatic joint, and an approximate universal joint (e.g., a 3-UPU (universal-prismatic-universal) manipulator) when coupled with a respective number of capture cones in space (e.g., in situ) during a docking procedure. In some embodiments, the 3-UPU may comprise combination of a universal joint, a prismatic joint, and an approximate universal joint, for example, a spherical joint.

[0051] After initial capture, the lance motor **112** may retract the probe assembly **102** back toward the capture vehicle **10**. The lance motor **112** may force the docking cone **114** into contact with the docking assembly **200** (e.g., within the outer cone **202**) to further secure the target vehicle **11**. A rim **208** of the docking assembly **200** may be forced into the backstop plate **118**. The latches **120** may be actuated to engage with the rim **208** of the docking assembly **200** to further secure the target vehicle **11** into a rigidized connection.

[0052] To release the target vehicle **11** in a nondestructive manner (e.g., a repeatable manner), the vehicle capture assembly **100** may release the docking assembly **200** and return to the initial position. For example, the lance motor **112** may extend the probe assembly **102** back away from the capture vehicle **10**. As discussed above, force applied to the probe tip **108** may overcome the biasing element **124** and move the probe tip **108** relative to the cam actuator **122**. The cam tip **126** of the cam actuator **122** may rotate the barbs **106** to release the docking assembly **200** and the probe tip **108** may be removed from the docking assembly **200**. Once the force is removed from the probe tip **108**, the biasing element **124** may return the probe tip **108** to the extended position where the barbs **106** may be returned to a deployed or capture position. The lance motor **112** may return the probe assembly **102** to the initial position similar to that shown in FIGS. 2 and 3.

[0053] To release the target vehicle **11** in another manner, one or more of the vehicle capture assembly **100** and/or the docking assembly **200** may include a destructively or nondestructively releasably union. For example, a coupling portion **210** may be coupled to the target vehicle **11** and releasably coupled to the remaining portion of the docking assembly **200** via a releasable union **212**. In some embodiments, the releasable union **212** may include a pyrotechnic coupling (e.g., one or more exploding bolts) that may explosively release the docking assembly **200** from the target vehicle **11**. In additional embodiments, the releasable union **212** may be a nondestructively releasable union (e.g., a remotely releasably electronic and/or magnetic latch or coupling).

[0054] FIG. 6 is an isometric view of a vehicle capture system **300** including multiple vehicle capture assemblies **302**. In some embodiments, the multiple vehicle capture assemblies **302** may be similar to and include similar components of the vehicle capture assemblies discussed above.

[0055] As shown in FIG. 6, the vehicle capture assemblies **302** (e.g., three assemblies or arms) may be coupled in a staggered formation (e.g., in a ring) on a capture vehicle **304** (e.g., which may be similar to the capture vehicle **10** (FIG. 1)). As depicted, the vehicle capture assemblies **302** may be offset from a central portion or centerline of the capture vehicle **304**. Such offset may provide clearance for propulsion systems or elements (e.g., primary engines) of the capture vehicle **304** and/or the target vehicle.

[0056] Two or more of the vehicle capture assemblies **302** may collectively (e.g., substantially simultaneously) dock with the target vehicle and draw and secure the target vehicle and capture vehicle **304** together by substantially simultaneously retracting each lance assembly of the vehicle capture assemblies **302**. In some embodiments, the vehicle capture assemblies **302** may provide redundant docking connections in case of a misaligned or otherwise failed docking with one or more of the vehicle capture assemblies **302**.

[0057] FIG. 7 is a partial cross-sectional side view of a vehicle capture assembly (e.g., vehicle capture assembly **400**) in an extended position (e.g., an initial position and/or a deployed position). In some embodiments, the vehicle capture assembly **400** may be similar to and include one or more components of the vehicle capture assembly **100** discussed above.

[0058] As shown in FIG. 7, the vehicle capture assembly **400** includes an actuation feature (e.g., lanyard actuator **422**) that may be coupled to and/or move within the lance assembly **404**. The lanyard actuator **422** may move (e.g., translate) relative to one or more portions of the probe assembly **402** (e.g., the probe tip **408**). For example, the lanyard actuator **422** may translate relative to (e.g., slide within) the probe tip **408**. As depicted, the lanyard actuator **422** may be coupled to one or more barbs **406** (e.g., via linkages **410**) in order to move the barbs **406** between extended and retracted positions (e.g., where each barb **406** is connected to a single, common lanyard actuator **422**). For example, as the lanyard actuator **422** translates within the probe assembly **402**, the lanyard actuator **422** may actuate the barbs **406** between positions. As depicted, the linkages **410** may be substantially J-shaped or L-shaped to provide leverage for rotating the barbs **406** and/or to provide adequate clearance for the lateral ends of the barbs **406**.

[0059] In some embodiments, the one or more barbs **406** may be coupled together. For example, the one or more barbs **406** may be coupled by pin **412** where the barbs **406** may rotate relative to one another and relative to the probe tip **408**. In some embodiments, the pin **412** may move (e.g., translate) within the probe tip **408** to enable the barbs **406** to rotate a selected amount (e.g., a selected range of degrees) in order to release and/or capture a target spacecraft. In additional embodiments, a portion of the probe tip **408** may move (e.g., translate) to enable the barbs **406** to rotate a selected amount in order to release and/or capture a target spacecraft.

[0060] In some embodiments, each of the barbs **406** may rotate (e.g., pivot) about a component of the probe tip **408** (e.g., one or more cam rollers **418**). As noted above, where the pin **412** is implemented, movement of the pin **412** and/or movement of the cam rollers **418** (e.g., with the probe tip **408**) may enable the rotation about of selected range of degrees. For example, the configuration may enable the barbs **406** to rotate **90** degrees to **180** degrees between retracted, extended, and overextended positions of each of the barbs **406**, as discussed in further detail below.

[0061] As depicted, a biasing element **424** (e.g., a spring) may be positioned between the lanyard actuator **422** and the probe tip **408** to bias the lanyard actuator **422** and/or the probe tip **408** in a selected position. For example, the biasing element **424** may bias the probe tip **408** in a position where the lanyard actuator **422** positions the barbs **406** in an extended or deployed position with the linkages **410** (e.g., where the barbs **406** extend laterally from the probe tip **408** at a substantially maximum dimension or width). When a force (e.g., force applied to the barbs **406** during a docking or undocking procedure) overcomes the biasing element **424**, the lanyard actuator **422** may move to compress and/or extend the biasing element **424** and, after the force is removed, the barbs **406** and lanyard actuator **422** may return to the initial state, as discussed below.

[0062] In some embodiments, the vehicle capture assembly **400** is coupled to a lance boom (e.g., the lance boom **110** (FIG. 4)) where extension (e.g., substantially full extension) of the lance boom toward a target spacecraft may act to pull the lanyard actuator **422** downward (e.g., against the biasing member **424**) in order to retract the barbs **406**. Such a configuration may enable retraction of the barbs **406** passively (e.g., without the direct use of a motor or other active device on the barbs or linkages coupled thereto). For example, the barbs **406** may be retracted passively using only the indirect motion of the lance boom, which may be driven by a motor, where the barbs **406** (or linkages coupled to the barbs **406**) are not directly driven by a motor or other active device.

[0063] In some embodiments, an end (e.g., a barb stow slider) of the lance boom that is movable relative to the length of the boom (e.g., and may be biased relative to the length of the lance boom) may be coupled to the lanyard actuator **422** by a component, such as a lanyard running through the lance boom. Once the lance boom approaches or substantially reaches the end of the stroke of the lance boom, the length of the lance boom may continue to move relative to the barb stow slider

(e.g., against a biasing force). The barb stow slider may then act to pull the lanyard actuator **422** in the opposite direction of the lance boom movement in order to retract the barbs **406**.

[0064] Similar to that discussed above, the vehicle capture assembly **400** may include another rearward engaging and/or retention element (e.g., docking cone **414**) for engaging another portion of the target vehicle **11** (e.g., another portion of the docking element **18** (FIG. 1)). As depicted, docking cone **414** may be biased toward the barbs **406** (e.g., by spring **416**) in order to secure the target vehicle between the barbs **406** and the docking cone **414**.

[0065] FIGS. 8 and 9 show the vehicle capture assembly **400** in use (e.g., during a docking procedure). As shown in FIG. 8, during the docking procedure, a portion of the target spacecraft **11** (e.g., a throat **430** of a docking element **18**) may force the barbs **406** into another position (e.g., the stowed or release position) from the initial position shown in FIG. 7 where the lateral width of the barbs **406** is decreased. With the barbs **406** at least partially retracted, the probe tip **408** and the barbs **406** may be positioned within the docking cone **414** and may travel past a necked portion of the docking element **18** (e.g., the throat **430**). Once the force is removed from the probe tip **408**, the biasing element **424** may return the probe tip **408** to an extended position where the barbs **406** may be returned to a deployed or capture position (e.g., in order to secure the target vehicle **11** via the docking element **18**). This renewed deployed or capture position may be similar to the expended position shown in FIG. 7 that creates a maximum lateral span of the barbs **406**. In the deployed or capture position, the barbs **406** may secure the vehicle capture assembly **400** to the docking element **18** of the target vehicle **11** (e.g., where the barbs **406** prevent travel of the probe tip **408** back through the throat **430**).

[0066] As shown in FIG. 9, after or during the docking procedure, the barbs **406** may be capable of moving to yet another release position to release the docking element **18**. Such an additional release position may be utilized to free the target vehicle **11** by releasing the docking element **18** from the probe tip **408** during an undocking procedure or during an unsuccessful or partial docking procedure. As depicted, the docking element **18** (e.g., the throat **430**) may force the barbs **406** into the additional release position (e.g., a forward or overextended position where the barbs **406** are pointed toward the docking element **18** and/or a distal end of the probe tip **408**). For example, relative movement between the vehicle capture assembly **400** and the target vehicle **11** may again force the barbs **406** into the surfaces of the docking element **18** defining the throat **430**.

[0067] In some embodiments, in order to move to the forward position, the barbs **406** may pivot against the cam rollers **418** and the pin **412** may move away from the cam rollers **418** to enable the barbs **406** to retract into the forward position as shown in FIG. 9. As discussed above, the pin **412** and/or the cam rollers **418** (e.g., via movement of the probe tip **408**) may translate to enable this rotation of the barbs **406**.

[0068] In the forward position, the docking element **18** may be released from the barbs **406** as the lateral width of the barbs **406** is decreased as the barbs **406** rotate forward. As noted above, such a release may be utilized when during an undocking procedure or when a docking procedure is unsuccessful, enabling the target vehicle **11** to be released.

[0069] In some embodiments, the amount of force required to move the barbs **406** to the forward position, may be selected to be greater than an expected range of forces experience during a normal docking procedure and may be greater than the force required to move the barbs **406** to the retracted position. For example, about 25 pounds of force (about 111.2 N) may be required to move the barbs **406** to the forward position and 2 pounds of force (about 8.9 N) may be required to move the barbs **406** to the retracted position. In such an embodiment, creating the relatively larger force between the vehicle capture assembly **400** and the target vehicle **11** may enable a passive undocking or release procedure between the vehicle capture assembly **400** and the target vehicle **11**.

[0070] As noted above, the barbs **406** may rotate up to **180** degrees from the retracted position or up to 90 degrees from the extended position in the forward position.

[0071] Once the force is removed from the probe tip **408**, the probe tip **408** may return the barbs **406** to the extended position from the overextended position where the barbs **406** may be returned to the deployed or capture position (e.g., in order to again secure the target vehicle **11** or another vehicle). In some embodiments, the biasing element **424** and the lanyard actuator **422** may act to force the barbs **406** back to the initial position or another biasing or force feature may be utilized to return the barbs **406** from the overextended position.

[0072] As discussed above, the movement of the barbs **406** may be accomplished passively, for example, where biasing forces from one or more biasing features are overcome to move the barbs **406** to either the retracted or forward positions. The biasing forces may also be used to return the barbs **406** into the initial position in the event the barbs **406** are moved (e.g., rotated) by another force. In additional embodiments, the movement of the barbs **406** may be accomplished by actively driving the barbs **406** (e.g., via the lanyard actuator **422** and/or linkages **410**) between one or more of the retracted or forwarded positions (e.g., via a motor of the vehicle capture assembly **400**).

[0073] In some embodiments, and similar to that discussed above, a force applied to the probe tip **408** may move the barbs **406** and the lanyard actuator **422** such that the barbs **406** release the docking assembly **18**. For example, a force applied to the probe tip **408** may force the probe tip **408** into the barbs **406**. In response, the barbs **406** (e.g., under the biasing force applied by the lanyard actuator **422**) may be rotated into the forward position such as that shown in FIG. **9** in order to release the docking element **18**.

[0074] The embodiments of the disclosure described above and illustrated in the accompanying drawing figures do not limit the scope of the disclosure, since these embodiments are merely examples of embodiments of the disclosure. The disclosure is defined by the appended claims and their legal equivalents. Any equivalent embodiments lie within the scope of this disclosure. Indeed, various modifications of the present disclosure, in addition to those shown and described herein, such as alternative useful combinations of the elements described, will become apparent to those of ordinary skill in the art from the description. Such modifications and embodiments also fall within the scope of the appended claims and their legal equivalents. The terminology used herein was chosen to explain the principles of the embodiments, the practical application or technical improvement over technologies found in the marketplace, or to enable others of ordinary skill in the art to understand the embodiments disclosed herein.

Claims

1. A vehicle capture assembly, comprising: a probe assembly including one or more retention elements for engaging with and securing a target vehicle in a deployed position, the probe assembly configured to passively engage with and secure the target vehicle with the one or more retention elements in the deployed position; a biasing element for biasing the one or more retention elements in the deployed position, the one or more retention elements configured to be moved toward a stowed position against a force of the biasing element in response to a force applied to the one or more retention elements; an extendable lance coupled to the probe assembly, the probe assembly being positioned at a distal portion of the lance; and an actuation element coupled to the one or more retention elements by one or more linkages, the actuation element for interacting with the biasing element to bias the one or more retention elements in the deployed position and to return the one or more retention elements to the deployed position after being forced into the stowed position.
2. The vehicle capture assembly of claim 1, wherein translation of the actuation element away from the one or more retention elements is configured to move the one or more retention elements into the stowed position in order to release the target vehicle.
3. The vehicle capture assembly of claim 1, wherein the one or more retention elements are configured to move past the deployed position and away from the stowed position in a direction

toward the target vehicle to an overextended position where the one or more retention elements release the target vehicle.

4. The vehicle capture assembly of claim 1, wherein translation of the lance in a direction toward the target vehicle is configured to move the one or more retention elements into the stowed position in order to release the target vehicle.
5. The vehicle capture assembly of claim 1, wherein the one or more retention elements are biased in the deployed position.
6. The vehicle capture assembly of claim 5, wherein the one or more retention elements are configured to move past the deployed position in a direction toward the target vehicle to an overextended position where the one or more retention elements release the target vehicle.
7. The vehicle capture assembly of claim 1, further comprising at least one clamp configured to engage with a docking cone of the target vehicle after the target vehicle has been secured by the probe assembly.
8. The vehicle capture assembly of claim 7, wherein, after engagement, the at least one clamp comprises a primary connection between the target vehicle and the vehicle capture assembly.
9. The vehicle capture assembly of claim 1, wherein the vehicle capture assembly comprises two or more capture arms, each of the two or more capture arms comprising a respective combination of a probe assembly and an extendable lance that is at least partially independent from the probe assembly and the extendable lance of another capture arm of the two or more capture arms.
10. The vehicle capture assembly of claim 9, wherein the two or more capture arms are each configured to be mounted at a location on a capture vehicle that is offset from a centerline of the capture vehicle and from primary propulsion devices of the capture vehicle.
11. A spacecraft capture system, comprising: two or more vehicle capture assemblies, each comprising: a probe assembly including one or more retention elements for engaging with and securing a target spacecraft; and an extendable lance coupled to the probe assembly, the probe assembly being positioned at a distal portion of the lance, wherein the probe assembly is configured to engage with a portion of the target spacecraft; wherein the two or more vehicle capture assemblies are configured to substantially simultaneously retract each respective probe assembly of the two or more vehicle capture assemblies in order to secure the target spacecraft.
12. The spacecraft capture system of claim 11, wherein each of the two or more vehicle capture assemblies are configured to substantially define a universal joint between the target spacecraft and the capture system when the probe assembly of each of the two or more vehicle capture assemblies is received in and engaged with a respective docking cone of the target spacecraft.
13. The spacecraft capture system of claim 11, wherein the probe assembly the extendable lance of each of the two or more vehicle capture assemblies are partially independent from the probe assembly and the extendable lance of another of the two or more vehicle capture assemblies.
14. The spacecraft capture system of claim 11, wherein the two or more vehicle capture assemblies are each configured to be mounted at a location on a capture vehicle that is offset from a centerline of the capture vehicle and from primary propulsion devices of the capture vehicle.
15. A method of capturing a target spacecraft, the method comprising: biasing one or more barbs of a probe of a vehicle capture assembly in a deployed position with an actuation element coupled to the one or more barbs; inserting the one or more barbs into a docking element of the target spacecraft; at least partially retracting the one or more barbs to a retracted position in response to a force applied to the one or more barbs by the target spacecraft; moving the one or more barbs back to the deployed position to secure the one or more barbs to the docking element with the actuation element; moving the actuation element along the probe against a biasing force; in response to moving the actuation element, moving the one or more barbs to a release position; and disengaging the one or more barbs from the docking element of the target spacecraft.
16. The method of claim 15, wherein moving the one or more barbs to the release position comprises forcing the one or more barbs in a direction toward the retracted position.

- 17.** The method of claim 15, wherein moving the one or more barbs to the release position comprises forcing the one or more barbs in a direction away from the retracted position and toward the target spacecraft with a force applied to the one or more barbs with the target spacecraft.
- 18.** The method of claim 15, further comprising destructively releasing the target spacecraft by detonating one or more explosive unions to disengage the target spacecraft from the probe.
- 19.** The method of claim 15, further comprising nondestructively releasing the target spacecraft to disengage the target spacecraft from the probe.
- 20.** The method of claim 19, wherein nondestructively releasing the target spacecraft comprises rotating the one or more barbs past the deployed position in a direction toward the target spacecraft.
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